INCH-POUND

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13 December 1988
SUPERSEDING
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MILITARY SPECIFICATION

DISINFECTANT, FOOD SERVICE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1.1 Scope. This document covers two types of disinfectants for use in food service preparation and utensils (see 6.1).

1.2 Classification. The disinfectant shall be of the following types, as specified (see 6.2).

Type I - Disinfectant based on trichloromelamine and potassium iodide

Type II - Disinfectant based on potassium dichloro-s-triazine trione and potassium iodide

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 6840

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SPECIFICATIONS

FEDERAL

RR-S-366	-	Sieve Test
PPP-B-26	-	Bag, Plastic (General Purpose)
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-676	-	Boxes, Setup

MILITARY

MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible

STANDARDS

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY

40 CFR 162 - Regulations for the Enforcement of the Federal Insecticide, Fungicide and Rodenticide Act

(The code of Federal Regulations (CFR) and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Reprints of certain regulations and application forms for registration of economic poisons may be obtained from the Pesticide Regulations Division, Environmental Protection Agency, 401 M Street SW, Washington, DC 20460.)

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* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

SOAP AND DETERGENT ASSOCIATION

ABS and LAS Biodegradability Scientific and Technical Report No. 3

(Copies may be obtained from the Soap and Detergent Association, 475 Park Avenue, New York, NY 10016.)

Food Chemicals Codex, 3rd ed. 1981

(Copies may be obtained from the Soap and Detergent Association, 485 Madison Avenue, New York, NY 10022.)

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP) and the National Formulary (NF)

(Copies may be obtained from the United States Pharmacopeia Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D 3951 - Standard Practice for Commercial Packaging

(Copies may be obtained from the American Society of Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* (Nongovernment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Laboratory report approval. Unless otherwise specified (see 6.2) at the time a bid is submitted, the contractor shall submit to the contracting officer a certified copy of a current year laboratory report covering the quality and grade of materials that he proposes to deliver. The laboratory report shall contain test data demonstrating that the food service disinfectant proposed for delivery by the contractor has been tested in accordance with the tests in section 4 and found to comply with the requirements of this document. Any one of the following types of reports will be satisfactory for this requirement:

a. An independent or commercial laboratory report
b. The prospective contractor's own laboratory report
c. A Governmental laboratory report from a contract

The purpose of this requirement is to assist in determining the capability of contractors to conform to all requirements of this document. The submission of an acceptable report under this requirement shall not be construed as relieving a contractor from subsequently meeting all requirements of this document on all deliveries.

3.2 Material. The food service disinfectant shall consist only of the following materials as applicable and described in 3.2.1 and 3.2.2.

3.2.1 Type I food service disinfectant. Type I food service disinfectant consists of trichloromelamine, a surface active material, anhydrous citric acid conforming to the requirements of U.S. Pharmocopeia, anyhdrous mono sodium dihydrogen phosphate and potassium iodide (see 6.4).

3.2.2 Type II food service disinfectant. Type II food service disinfectant consists of potassium dichloro-s-triazine trione, a surface active material, anhydrous tartaric acid conforming to the requirements of Food Chemicals Index, anhydrous mono sodium dihydrogen phosphate, and potassium iodide (see 6.4).

3.2.2.1 Surface active material. The surface active material shall be white to a light cream in color, anionic type, and the active ingredient which consists of linear sodium alkyl (C_{11}-C_{12}) aryl sulfonate (LAS). The surface active material shall be at least 80 percent biodegradable when tested as specified in 4.4.1.1.

3.2.2.2 Iodine uptake. The iodine uptake shall be no more than 4.5 milligrams (mg) of iodine per gram of surface active material when tested as specified in 4.4.1.2.

3.2.2.3 Active ingredient. The surface active material shall have an active ingredient content of not less than 81 percent when tested as specified in 4.4.1.3. (Commercially available surfactants suitable for use in this item are generally 81 to 87 percent active.)

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3.2.2.4 Moisture content. The moisture content shall be no greater than 1.25 percent when tested as specified in 4.4.1.4.

3.2.3 Potassium iodide. The potassium iodide shall be in granular form and shall conform to the requirements of the United States Pharmacopeia.

3.3 Composition. When tested as specified in 4.3.6, the food service disinfectant shall conform to the requirements given below.

3.3.1 Type I. The standard trichloromelamine formula is specified in table I. In addition, the disinfectant requires the addition of a potassium iodide pouch, packaged separately, to the standard trichloromelamine pouch.

TABLE I. Standard trichloromelamine formula

Ingredient	Percent by weight Minimum	Maximum
Trichloromelamine	26.9	27.9
Anionic linear sodium alkyl aryl sulfonate surface active material, technical (LAS)	11.5	12.5
Citric acid, anhydrous	51.0	55.0
Monosodium dihydrogen phosphate, anhydrous	Remainder	Remainder

3.3.1.1 4.77 ounces trichloromelamine-potassium iodide pouches. Unless otherwise specified (see 6.2), each trichloromelamine pouch shall contain 3.43 ± 0.10 ounces of the standard trichloromelamine formulation and each potassium iodide pouch shall contain 1.34 ± 0.10 ounces of potassium iodide. The pouches, prepared as separate units, shall be constructed of heat sealable barrier material consisting of the following lamination of materials (from inside to outside surface): 0.0015 inch polyethylene by 0.002 inch aluminum foil by 0.0015 inch polyethylene by 25 pound basis weight pouch paper with 2 pound basis weight pouch paper with 2-pound basis weight per ream (24 by 36-500) lacquer coating (see 6.6). Pouches shall be one or two piece construction. Side, top and bottom seams of the two-piece pouch shall be heat sealed with the seams a minimum of 3/16 inch in width. In addition to the minimum of 3/16 inch seal on the top, a lip of at least 1/2 inch shall be present at the top of each pouch for joining and heat sealing both the trichloromelamine and the potassium iodide pouches together. After each pouch has been filled and closure heat sealed, one trichloromelamine pouch labeled pouch "A" and one potassium iodide pouch labeled pouch "B" shall be heat sealed together by joining the top 1/2 inch lip of each pouch. In addition, a tearing notch shall be positioned on the top seam of each

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trichloromelamine and potassium iodide pouch. The notch in each pouch shall be located in the top seam so that both pouches may be opened in one operation to ensure delivery of the contents during use. One-piece pouch seams and closure shall be as specified in 3.3.3 and 3.3.4. When tested as specified in 4.4.3.1 and 4.4.3.2, the finished pouches shall show no signs of visual deterioration. The exterior of the pouches shall be free from visible dust. The joined pouches shall be enclosed in a close-fitting heat sealed bag, fabricated from material conforming to type II, grade A, class 1 of MIL-B-121.

3.3.2 Type II. The standard potassium dichloro-s-triazine trione formula is specified in table II. The disinfectant requires the addition of a potassium iodide pouch, packaged separately, to the standard potassium dichloro-s-triazine trione pouch.

3.3.2.1 4.77 ounces chlorinated cyanurate-potassium iodide pouches. Unless otherwise specified (see 6.2), each chlorinated cyanurate pouch shall contain 3.43 ± 0.10 ounces of the standard chlorinated cyanurate formulation and each potassium iodide pouch shall contain 1.34 ± 0.10 ounces of potassium iodide. The pouches, prepared as separate units shall be constructed of heat sealable barrier material consisting of the following lamination of materials (from inside to outside surface) 0.0015 inch polyethylene by 0.002 inch aluminum foil by 0.0015 inch polyethylene by 25 pound basis weight pouch paper with 2 pound basis weight per ream (24 by 36-500) lacquer coating (see 6.6). Pouches shall be one or two piece construction. Side, top and bottom seams of the two-piece pouch shall be heat sealed with the seals a minimum of three-sixteenths of an inch in width. In addition to a minimum of three-sixteenths inch seal on the top, a lip of at least 1/2 inch shall be present at the top of each pouch for joining and heat sealing both the chlorinated cyanurate and the potassium iodide pouch labeled pouch "B". The pouches shall be heat sealed together by joining the top 1/2 inch lip of each pouch. In addition, a tearing notch shall be positioned on top seam of each chlorinated cyanurate and potassium iodide pouch. The notch in each pouch shall be located in the top seal so that both pouches may be opened in one operation to insure delivery of the contents during use. One piece pouch seams and closure shall be as specified in 3.3.3 and 3.3.4. When tested as specified in 4.4.3.1 and 4.4.3.2, the finished pouches shall show no signs of visual deterioration. The exterior of the pouches shall be free from visible dust. The joined pouches shall be enclosed in a close-fitting heat sealed bag fabricated from material conforming to type II, grade A, class 1 of MIL-B-121.

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TABLE II. Standard Chlorinated Cyanurate Formula

Ingredient	Percent by weight Minimum	Maximum
Potassium dichloro-s-triazine trione	40.50	44.5
Anionic linear sodium alkyl aryl sulfonate surface active material, technical (LAS)	11.5	12.0
Tartaric acid, anhydrous	43.0	43.5
Monosodium dihydrogen phosphate, anhydrous	Remainder	Remainder

3.3.3 Pouch side seams and closure. All side seams and closures shall be heat sealed with the seal being a minimum of 3/16 inch in width.

3.3.4 Bottom fold sealing. The bottom fold-over shall be heat sealed with the seal being a minimum of 1/16 inch in width.

3.3.5 Leakage of seams and closure of pouch. The seams and closure of the pouch shall be complete and continuous and shall not show signs of leakage or bubbling when immersed in water, or shall not wet the contents when tested as specified in 4.4.3.3.

3.4 Disinfectant pouches, finished product. The finished product consists of pouch "A" and pouch "B" as follows:

Type I, pouch A - contains trichloromelamine formula (see table I)

Type II, pouch A - contains potassium dichloro-s-triazine trione formula (see table II)

Pouch B - contains potassium iodide (for types I and II)

3.4.1 Titratable chlorine. The contents of the pouch A (type I or type II) shall contain not less than 24.0 percent nor more than 26.5 percent of titratable chlorine when tested as specified in 4.4.2.1.

3.4.2 Acidity. The contents of the pouch A (type I or type II) shall have a pH value not less than 2.5 and shall have a titratable acidity equivalent to not less than 200 mg of sodium hydroxide per gram of product when tested as specified in 4.4.2.2.

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3.4.3 Surface active ingredient. The contents of the pouch A (type I or type II) shall have not less than 9.1 percent nor more than 10.7 percent of active ingredient when tested as specified in 4.4.2.3. (These values are slightly less than theoretical for there is seldom complete conversion in the test method in 4.4.2.3.)

3.4.4 Solubility. The contents of the pouch A (type I and type II) shall be soluble when tested as specified in 4.4.2.4. The titratable chlorine in the sample shall not be less than 152 mg per liter.

3.4.5 Moisture. The content of the pouch A (type I and type II) shall have a moisture content of not more than 0.60 percent when tested as specified in 4.4.2.5.

3.4.6 Particle size. Not less than 95 percent, by weight, of the content of the pouch A (type I and type II) shall pass through a No. 20 standard sieve conforming to RR-S-366, and not less than 100 percent through a No. 10 standard sieve conforming to RR-S-366, when tested as specified in 4.4.2.6.

3.4.7 Bactericidal performance. When tested as specified in 4.4.2.7, the contents of the pouch A shall give a 99.999 percent kill in 90 seconds for at least 2 of the 3 tests.

3.4.8 Formulation stability. The contents of the pouch A shall not lose more than 20 percent of the original amount of titratable chlorine when determined as specified in 4.4.2.8.

3.4.9 Surface tension. A solution of the contents of the pouch A shall have a surface tension of not more than 40 dynes per centimeter when prepared and tested as specified in 4.4.2.9.

3.5 Labeling. The labeling shall be registered under the Federal Insecticide, Fungicide, and Rodenticide Act and be in compliance with its requirements. The contractor shall use EPA registration numbers (given below) assigned to the US Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014. The contractor shall use the EPA establishement number assigned to his/her manufacturing plant.

Type I - EPA registration number is 40510-1. The EPA approved labels shall be as shown in figures 1 and 2. In addition to the marking specified in 5.4, the information found in figures 1 and 2 shall be legibly marked on each bag (containing the double compartmented pouch) and each interior box.

Type II - EPA registration number is 40510-U. The EPA approved labels shall be as shown in figures 3 and 4. In addition to the marking specified in 5.4, the information found in figures 3 and 4, shall be legibly marked on each bag (containing the double compartmented pouch) and each interior box.

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3.5.1 Additional marking. In addition to the above marking, the following information shall be legibly printed on both sides of each appropriate pouch:

Type I - Trichloromelamine pouch A
Potassium iodide pouch B

Type II - Chlorinated cyanurate pouch A
Potassium iodide pouch B

3.5.2 Instruction sheet. A sheet of chemical wood writing paper, legibly printed with the following instruction, shall be inserted in each interior box specified in 5.1.1:

INSTRUCTION FOR DETERMINATION OF TITRATABLE CHLORINE WHEN DISINFECTANT, FOOD SERVICE IS TO BE ISSUED AFTER _____ (the same date stamped on caution label affixed to shipping container shall be inserted here).

Weigh out 1 gram of the contents of the pouch A (trichloromelamine pouch for type I or chlorinated cyanurate pouch for type II) to the nearest milligram into a round bottom test tube (approximately 14 millimeters (mm) in diameter and 160mm in length). Add 0.5 milliliters (mL) of distilled water to the sample, and stir with a round tipped stirring rod (of a size convenient to fit into tube). Mix the sample into a smooth paste for exactly 3 minutes. Completely transfer the paste by rinsing into a 1.5 liter beaker, using a total of exactly 1 liter of distilled water. Stir the solution for exactly 30 seconds with a large stirring rod (do not whip or splash when stirring) and then let stand for 2 minutes. At the end of this time, insert a 20 mL pipette 1 inch below the surface and draw 20 mL into the pipette. Place the 20 mL sample into a 125 mL Erlenmeyer flask or iodine flask and analyze for titratable chlorine by the following procedure:

Dissolve 1 gram of potassium iodide crystals into the 20 mL sample. Add six drops of 50 percent acetic acid and swirl to ensure complete mixing. Allow to stand five minutes in the dark. Titrate the free iodine at once with freshly standardized 0.005 N sodium thiosulfate. When the color is straw yellow, add starch indicator and titrate to end point. Analysis should be conducted in triplicate and results reported as an average.

3.5.2.1 Calculation. Calculate the results as follows.

$$\text{Percent titratable chlorine} = \frac{(35.45)\ (\text{mL of sodium thiosulfate})\ (N)\ (50)\ (100)}{(\text{gram sample})\ (1000)}$$

Where N = normality of sodium thiosulfate solution.

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3.6 Workmanship. The food service disinfectant shall be a noncaking, clean, uniform powder. The entire process of mixing, grinding (see 6.3 and 6.5), filling and sealing of the individual pouches shall be carried out in surroundings maintained at a relative humidity of not more than 45 percent and at a temperature of not more than 80°F, and in such a manner as to ensure a product of consistently high quality. All components and bulk mixed finished formulation shall be unit packed in moisture-vapor proof containers and kept dry and protected from moisture until incorporated into final packaging.

3.6.1 Additional workmanship. The potassium iodide cannot be mixed intimately with the rest of the pouch A (type I or type II) formulation. The unit packing of the potassium iodide and the pouch A (type I or type II) finished product shall be carried out in separate rooms under the controlled humidity and temperature conditions specified in 3.6. A two-compartmented pouch type package shall be used to keep the potassium iodide separated from the Pouch A formulation until time for use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

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4.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.3.1.1 Component testing. Component testing shall be performed in accordance with table III. Each lot of surface active material shall be checked for all requirements prior to incorporation into formulation. For purposes of sampling, a lot shall consist of all the surface active material received from one manufacturer, produced at one plant from the same raw materials and under essentially the same manufacturing conditions. The lot size shall be expressed in pounds of surface active material. The sample unit for testing, except color, shall be a one-half pound composite obtained by randomly selecting and thoroughly mixing one-quarter pound samples in the frequency indicated below. The composite shall be placed in a clean, dry, sealed container of the smallest possible size. Care shall be exercised to prevent contamination or alteration of the composite during the sampling, weighing and storage. The lot shall be unacceptable if the composite fails to meet any of the test requirements specified. All test reports shall contain the individual values utilized in expressing the final result.

Lot size (pounds)	Frequency
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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TABLE III. Instructions for testing of components

Characteristic	Document reference		Requirement applied to		Number determinations per unit	Results reported as	
	Requirement	Test method	Ind. unit	Lot avg.		Pass or fail	Numerically to nearest
Surface active material							
Color	3.2.2.1	3.2.2.1	X1/			X	
Biodegradability	3.2.2.1	4.4.1.1		X	2		Average of 2 determinations to nearest 0.1 percent
Iodine uptake	3.2.2.2	4.4.1.2		X	2		Average of 2 determinations to nearest 0.5 mg
Active ingredient	3.2.2.3	4.4.1.3		X	2		Average of 2 determinations to nearest percent
Moisture	3.2.2.4	4.4.1.4		X	2		Average of 2 determinations to nearest 0.01 percent
Potassium iodide	3.2.3	2/					

1/ All containers shall be examined for color. Failure of the contents of a container to meet the color requirement shall be cause for rejection of the contents of that container.

2/ The contractor shall furnish a certificate of compliance covering this requirement.

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4.3.1.2 Certification. The contractor shall furnish a certificate of compliance for Government approval for the requirements specified in 3.2.3 and the composition requirements specified in tables I and II.

4.3.2 In-process inspection. Inspection shall be made of the following operations or assemblies to establish conformance to specified requirements. Whenever nonconformance is noted, correction shall be made to the items affected and to the operation. Items which cannot be corrected shall be removed from production.

a. Batch mix composition formulation complies with 3.3.1 and 3.3.2

b. Workmanship complies with 3.6

4.3.3 End item visual examination. The filled pouches shall be examined for the defects listed in table IV. The lot size shall be expressed in units of filled bags. The sample unit shall be one filled bag. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 0.65.

TABLE IV. End item visual defects

Examine	Defects
Construction	Outer bag not constructed of materials specified
	Outer bag not sealed
	Pouch A (trichloromelamine for type I or chlorinated cyanurate for type II) missing
	Pouch B (potassium iodide) missing
	Tearing notch missing from pouches
	Tearing notch not positioned as specified
	Not in accordance with contract requirements
	Seams and closures of pouches not completely heat sealed throughout entire length and width
	Not uniformly constructed
	Pouches not constructed of materials specified
	Any evidence of seepage; any unsound seam
	Any tear, cut, slit, split, cracks, blisters or holes (including pinholes)
Workmanship	Not clean; any evidence of dirt, dust or foreign matter
	Presence of food service disinfectant, or potassium iodide on exterior of pouches, as applicable
	Caking; not uniform powder

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TABLE IV. End item visual defects (cont'd)

Examine	Defects
Markings	Markings pertaining to the applicable information and instructions for end item use, missing, illegible, incomplete or do not conform to applicable legend specified in 3.5 Not printed in black or blue ink Printing ink faded, blurred or smeared

4.3.4 End item dimensional examination. The filled bags shall be examined for the defects listed in table V. The lot size shall be expressed in units of filled bags. The sample unit shall be one filled bag. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

TABLE V. End item dimensional defects

Examine	Defects
Size of seams of pouch:	
Two piece construction	Side, top and bottom seams less than 3/16 inch wide
One piece construction	Side seams and closure less than 3/16 inch wide Bottom fold seal less than 1/16 inch wide Less than 1/2 inch lip present for joining and sealing pouch A and pouch B

4.3.5 Examination of the filled pouches for net contents. The lot size shall be expressed in units of filled pouches. The sample unit shall be one filled pouch. The net contents shall not be less than as specified or indicated. The inspection level shall be S-2 and the AQL shall be 2.5, expressed as percent defective.

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4.3.6 End item testing. The methods of testing listed in table VI shall be followed. The lot size shall be expressed in units of bags. The sample unit for testing the pouch A contents shall be five individual 3.43 ounce pouches. The sample unit for packaging testing shall be one polyethylene bagged intermediate box. The number of sample units to be tested shall be as indicated below. When specified in table VI, tests shall be performed on a sample unit or composite basis. The lot shall be unacceptable if a sample unit or the composite fails to meet any test requirement specified. All test reports shall contain the individual values utilized in expressing the final results.

Lot size (bags or pouches)	Number of units
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

4.3.6.1 Composite sample. One 3.43 ounce pouch shall be selected from each sample unit. The contents of the pouches so selected shall be thoroughly mixed to yield a homogeneous composite. The composite shall be placed in an airtight and watertight brown glass container, covered and sealed to prevent any moisture change.

4.3.6.2 Sample unit basis. The remaining pouches shall be used for testing on a sample unit basis. Care shall be exercised to prevent any change in the compound due to moisture prior to testing.

4.3.6.3 Frequency of testing. Except when otherwise specified in table VI, the tests for each characteristic shall be performed on a lot by lot basis. Those tests designated by an asterisk shall be performed on the first lot and on every fifth lot thereafter.

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TABLE VI. Instructions for testing of the end item

Characteristic	Document reference		Requirement applied to		Number determinations per unit	Results reported as	
	Requirement	Test method	Ind. unit	Lot avg.		Pass or fail	Numerically to nearest
Composition	Table I or Table II	1/					
Contents of pouch A							
Titratable chlorine	3.4.1	4.4.2.1	X		2		Average of 2 determinations to nearest 0.1 percent
Acidity	3.4.2						
pH value		4.4.2.2	X		1		0.1 pH
Titratable acidity		4.4.2.2	X		1		mg
Surface active ingredient	3.4.3	4.4.2.3		X	3		Average of 3 determinations to nearest 0.1 percent
Solubility	3.4.4	4.4.2.4		X	1	X	
Moisture	3.4.5	4.4.2.5	X		2		Average of 2 determinations to nearest 0.01 percent
Particle size	3.4.6						
Standard sieve No.20		4.4.2.6		X	1		0.1 percent/weight
Standard sieve No.10		4.4.2.6		X	1		0.1 percent/weight

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TABLE VI. Instructions for testing of the end item (cont'd)

Characteristic	Document reference		Requirement applied to		Number determinations per unit	Results reported as	
	Requirement	Test method	Ind. unit	Lot avg.		Pass or fail	Numerically to nearest
Bactericidal performance *	3.4.7	4.4.2.7		X	3		Average of 3 determinations to 0.001 percent kill
Formulation stability	3.4.8	4.4.2.8	X		1	X	
Surface tension	3.4.9	4.4.2.9		X	3		Average of 3 determinations to nearest dyne/cm
Pouches, filled and sealed:							
Finished pouch stability	3.3.1.1 3.3.2.1	4.4.3.1	X		1	X	
Stability after packaging	3.3.1.1 3.3.2.1	4.4.3.2	X		1	X	
Leakage of seams and closures	3.3.5	4.4.3.3	X		1	X	
Water spray test (interior package)	5.1.1	4.4.3.4	X		1	X	

1/ Certificate of compliance shall be furnished by the contracting officer for composition requirements except for surface active material.

* Tests performed on the first lot and/or every fifth lot thereafter.

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4.3.7 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling. Bulged or distorted container.
Content	Number per container is more or less than required.
Content	Number of intermediate containers is more or less than specified. Number of pouches per intermediate container is more or less than specified. 1/

1/ For this defect one intermediate container shall be examined.

4.3.8 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1, and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.

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Examine	Defect
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.4 Methods of inspection.

4.4.1 Tests on surface active material.

4.4.1.1 Biodegradability. The biodegradability of the surface active material shall be determined in accordance with the ABS and LAS Biodegradability Scientific and Technical Report No. 3.

4.4.1.2 Iodine uptake. Weigh a sample of 2 grams of the LAS surface active material to the nearest 0.005 gram on a piece of waxed paper. Transfer it to a 250 mL separatory funnel (Corning No. 6340 or equivalent with stem cut to approximately 0.5 inch length) containing 50 mL of hot distilled water. Without shaking, dilute to 75 mL with hot distilled water and swirl but do not shake to dissolve. Add 2 mL of starch indicator solution and titrate with 0.1 normal (N) iodine solution (12.692 grams iodine and 8 grams potassium iodide per liter of distilled water) until all sulfate is destroyed as indicated by the appearance of the starch-iodine color. Swirl during the titration and allow to stand with occasional swirling for 2 minutes after the initial appearance of the starch-iodine color to be certain that the color does not fade.

4.4.1.2.1 Calculation. Calculate the results as follows.

$$\text{Milligrams iodine} = \frac{(126.92)\ (\text{mL of iodine solution})\ (N)}{(\text{grams of sample})}$$

4.4.1.3 Active ingredient. Weigh out 2 grams of LAS to the nearest 0.1 mg in a plastic dish. Transfer to a 250 mL volumetric flask that is half full with distilled water. Mix well by shaking and dilute to volume 250 mL with distilled water. Then mix again. To a 250 mL stoppered graduated cylinder, add 25 mL of methylene blue indicator, 25 mL chloroform, 25 mL distilled water and pipette a 25 mL aliquot of LAS slowly into graduated cylinder. From a 50 mL buret add as large a volume as possible of .018N hyamine 1622 without risk of over titrating. Stopper the graduated cylinder and shake at least 30 times. To speed separation, hold the graduated cylinder in a horizontal position until most of the chloroform has separated, then return the graduated cylinder to the vertical position and swirl gently. After the mixture has separated, note color difference between the two layers. Using the degree of color difference as a guide, continue adding hyamine in successively smaller increments, shaking the graduated cylinder 30 times after each addition. When colors are close, add the hyamine in 0.05 mL amounts until end point, at which both will be aqua blue

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color. The end point is when the colors of the two layers match when the graduated cylinder is observed at eye level, against a white background. The comparison should be made as soon as the layers separate completely.

4.4.1.3.1 Calculation. Calculate the results as follows.

$$\text{Percent active ingredient} = \frac{(\text{mL hyamine})\ (\text{N hyamine})\ (\text{M.W.})\ (\text{dilution})}{(10)\ (\text{sample weight})\ (\text{aliquot})}$$

Where:

N = Normality of hyamine solution (see sections 4.4.1.3.5 and 4.4.1.3.6)
M.W. = Molecular weight of LAS (=340)
Dilution = Volume LAS made to (250 mL) of LAS solution
Aliquot = Volume of LAS solution used for titration (25 mL)

4.4.1.3.2 Reagents - hyamine 1662.

a. Hyamine 1622 crystals (benzethonium chloride), Aldrich Chemical Co., Milwaukee, WI 53201.
b. Sodium hydroxide, 50 percent solution, reagent grade
c. Sodium lauryl sulfate (dodecyl sulfate, sodium salt) crystals, Aldrich Chemical Co., Milwaukee, WI 53201

4.4.1.3.3 Reagents - methylene blue indicator.

a. Methylene blue chloride (652), Allied Chemical, National Aniline Division
b. Sulfuric acid, concentrated, reagent grade
c. Sodium sulfate, anhydrous, reagent grade

4.4.1.3.4 Preparation of hyamine 0.018N. Fill a one liter volumetric flask half full with distilled water. Then carefully add 3.20 grams of 50 percent sodium hydroxide and mix (in grams, because of the amount it is easier to weigh). Then add 7.36 grams of hyamine crystals and fill to volume with distilled water. Stir until dissolved. Do not dry the hyamine. The weight should be adjusted, if necessary, to provide a solution close to 0.18N.

4.4.1.3.5 Standardization of hyamine. Sodium lauryl sulfate (SLS) active solution; (by weight), .003N: Weigh approximately 0.873 grams of SLS to the nearest 0.1 mg (use standard as received, do not dry). Transfer to a one liter volumetric flask, dissolve in distilled water, dilute to volume and mix. Calculate the normality of SLS solution to six decimal places. When not being used, store the SLS in a tightly stoppered container. Label with normality, date and technician's initials. Discard after one year. To standardize hyamine, pipette a 20 mL aliquot from the 0.003N standard active SLS solution and add it to a 250 mL stoppered graduated cylinder already containing:

25 mL methylene blue indicator
25 mL chloroform
30 mL distilled water

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Titrate with the .002N hyamine solution using a 50 mL burette. As as large a volume as possible without risk of overtitrating. Stopper the graduated cylinder and shake vigorously at least 30 times. To speed up the separating, hold the graduated cylinder in a horizontal position until most of the chloroform has separated, then return the graduated cylinder to the vertical position and swirl gently. After the mixture has separated, note color difference between the two layers. Using the degree of color difference as a guide, continue adding hyamine in successively smaller increments, shaking the graduated cylinder 20 times after each addition. When colors of the two layers are close, add the hyamine in 0.5 mL amounts until the end point at which both layers will be an aqua blue color. The end point is when the colors of the two match when the graduated cylinder is observed at eye level, against a white background. The comparison should be made as soon as the layers separate completely.

4.4.1.3.6 Calculation. Calculate the results as follows.

$$\text{Normality of sodium lauryl sulfate (N SLS)} = \frac{\text{(weight of SLS) (percent purity)}}{\text{(100) (288.4)}}$$

$$\text{Where N hyamine} = \frac{\text{20 X (N SLS)}}{\text{mL hyamine}}$$

4.4.1.3.7 Labeling. Label the container with the normality, date and technician's initials.

4.4.1.3.8 Caution. Hyamine normality is affected by contact with certain types of plastics tubing. Therefore, it may be necessary to flush the tubing if solution has been standing two days or more.

4.4.1.3.9 Preparation of methylene blue indicator. Prepare a dye solution containing 3.0 grams of pure methylene blue per liter. The amount of dye depends on the percent dye stated on the container label and is calculated as follows:

$$\text{grams of dye} = \frac{\text{3.00 x percent dye content}}{100}$$

Wash the dye into a one liter volumetric flask and dilute to volume with distilled water. Mix well before each use. To prepare the methyl blue indicator, fill the container to be used half full with distilled water. Then for each liter of solution add: 10 mL of methylene blue concentrate, 8 mL of concentrated sulfuric acid and 50 grams of sodium sulfate. Mix well and dilute to volume. Mix well again. Label properly.

4.4.1.4 Moisture. Weigh sample of 3 grams of the LAS surface active material to the nearest 0.1 mg in a covered sample dish. Uncover the dish and dry the sample in an oven for 24 hours at 105° ± 2°C. Replace the cover, cool the sample to room temperature in a desiccator, and again weigh the sample to the nearest 0.1 mg.

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4.4.1.4.1 Calculation. Calculate the results as follows.

$$\text{Percent moisture} = \frac{(\text{loss of weight})\ (100)}{(\text{grams of sample})}$$

4.4.2 Tests on contents of pouch A (type I or type II).

4.4.2.1 Titratable chlorine. Thoroughly and quickly mix approximately 5 grams of the contents of the pouch A (type I or type II) sampled at random, by shaking in a closed glass container. Promptly obtain a representative sample of 1 gram from the 5 gram sample weighed to the nearest milligram, and place in a 500 mL Erlenmeyer flask. To this, add 2 grams of potassium iodide (KI) crystals and approximately 25 mL of glacial acetic acid. Stopper the flask and swirl for about 5 minutes to dissolve the sample. (Note: Carry out determination of titratable chlorine in subdued light.) Add approximately 25 mL of distilled water and again stopper and swirl to ensure complete mixing and solution of KI. Titrate at once with a freshly standardized 0.15 N sodium thiosulfate solution to light straw color. Add starch indicator, and continue titration to end point.

4.4.2.1.1 Calculation. Calculate the results as follows.

$$\text{Percent titratable chlorine} = \frac{(35.45)\ (\text{mL of sodium thiosulfate})\ (\text{N})\ (100)}{(\text{grams of sample})\ (1000)}$$

4.4.2.2 Acidity. Weigh 1 gram of the contents of the pouch A (type I or type II) to the nearest mg and disperse in 1 liter of distilled water at 25° ± 1°C. The pH, measured with a glass electrode, shall not be lower than 2.5. Titrate the entire solution with 0.1 N NaOH. The latter solution is added from a burette with stirring, measuring the pH with a glass electrode at suitable increments. As the pH value approaches 5.0, the increments should be made smaller, with pH intervals no greater than 0.2. Continue the titration until pH of 5.5 has been reached. The number of milliliters of 0.1 N NaOH required to give a pH of 5.0 is then found by interpolation of this data.

4.4.2.2.1 Calculation. Calculate the results as follows.

$$\text{Milligrams NaOH per gram} = \frac{(\text{mL of NaOH})\ (\text{N})\ (40)}{(\text{grams of sample})}$$

Where N = Normality of NaOH

4.4.2.3 Surface active ingredient. A 1.5 gram sample of disinfectant from pouch A (type I or type II) weighed out to the nearest mg shall be dispersed in a 250 mL volumetric flask containing 125 mL of distilled water. Mix well and fill to volume. Mix well again. Pipette a 25 mL aliquot of sample into a 125 mL flask. Add 2.0 N sodium arsenite dropwise. Test one drop of solution on starch iodine test paper until all blue disappears (available chlorine is removed). Record number of drops required. Discard. To a 250 mL stoppered graduated cylinder add 25 mL of methylene blue indicator, 25 mL chloroform, 25

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mL of distilled water and a 25 mL portion of sample, taken from the original flask while agitating. Add predetermined volume of sodium arsenite plus an excess of 1 or 2 drops as determined in the pretest. Titrate with standardized hyamine 1622 solution (.002N). Shake vigorously after each addition. After the mixture has separated, note color difference between the two layers. Using the degree of difference as a guide, continue adding hyamine in successfully smaller increments shaking the graduated cylinder two times after each addition. When colors are close, add the hyamine in 0.05 mL amounts until the end point.

4.4.2.3.1 Calculation. Calculate results as follows:

$$\text{Percent surface active ingredient} = \frac{(\text{mL hyamine})\ (\text{N hyamine})\ (\text{M.W.})\ (\text{dilution})}{(10)\ (\text{sample weight})\ (\text{aliquot})}$$

Where:

N = Normality of hyamine solution (see sections 4.4.1.3.5 and 4.4.1.3.6)
M.W. = Molecular weight of LAS (=340)
Dilution = Volume of sample (from pouch A) made up to (250 mL)
Aliquot = Volume sample solution used for titration (25 mL)

4.4.2.3.2 Preparation of hyamine 1622 (0.002N). Fill a one liter volumetric flask half full with distilled water. Add 0.4 grams 50 percent NaOH slowly and mix. Then add 0.82 grams of hyamine crystals. Stir until dissolved. Fill to volume with distilled water. For rest of standardizing procedures see 4.4.1.3.5.

4.4.2.3.3 Preparation of methylene blue. For preparation of methylene blue, see 4.4.1.3.9.

4.4.2.3.3.1 Regeants.

a. Sodium arsenite (2N) $NaAsO_2$, reagent grade, Mallinckrodt, St. Louis, MO
b. Sodium hydroxide solution 50 percent, reagent grade

4.4.2.3.4 Procedure. Wash 32.5 grams of sodium arsenite into a 250 mL volumetric flask, with distilled water. Add 19.7 mL of 50 percent sodium hydroxide, dilute to volume with distilled water and mix well, to give 2N solution. The solution is highly alkaline and susceptible to air oxidation. It should be stored in a plastic container and dispensed from a dropping bottle. NOTE: Promptly identify solution with a POISON label.

4.4.2.4 Solubility. Measure into a 1.5 liter beaker, a liter of distilled water at $25^{o} \pm 1^{o}C$. Weigh out 1 gram of the contents of the pouch A (type I or type II) to the nearest mg. Add carefully to the distilled water, and stir for 2 minutes with a glass rod. Stirring should not be so rapid as to cause splashing or whipping of the mixture but sufficient to keep the undissolved powder from

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settling to the bottom. Withdraw the stirring rod and allow the solution to stand 2 minutes. At the end of this time, insert a 20 mL pipette about 1 inch below the surface and draw 20 mL into the pipette. Place the 20 mL sample in a 125 mL Erlenmeyer flask and analyze for titratable chlorine by the following procedure: Dissolve 1 gram of KI crystals in the 20 mL sample. Add 6 drops of 50 percent acetic acid and swirl to ensure complete mixing. Allow to stand 5 minutes in the dark. Titrate the free iodine at once with freshly standardized 0.005 N sodium thiosulfate. When the color is straw yellow, add starch indicator and titrate to end point.

4.4.2.4.1 Calculation. Calculate the results as follows.

$$\text{Milligrams of titratable chlorine per liter} = \frac{(35.45)\ (\text{mL sodium thiosulfate})\ (N)\ (1000)}{(\text{mL of sample solution})}$$

Where N = Normality of thiosulfate solution.

4.4.2.5 Moisture. Weigh to the nearest 0.1 mg a sample of 3 grams of the contents of the pouch A (type I or type II) in a covered sample dish. Uncover the dish and dry the sample in a vacuum oven for 2 hours at 50° ± 2°C. Replace the cover, cool the sample to room temperature in a desiccator, and again weigh the sample to the nearest 0.1 mg.

4.4.2.5.1 Calculation. Calculate the results as follows.

$$\text{Percent moisture} = \frac{(\text{loss of weight})\ (100)}{(\text{grams of sample})}$$

4.4.2.6 Particle size. Weigh a representative 10 ± 0.05 gram sample of the contents of the pouch A (type I or type II) and place on a No. 10 standard sieve conforming to RR-S-366 that is fitted into a No. 20 standard sieve conforming to RR-S-366. The No. 20 sieve is fitted on the receiving pan. After introducing the sample, place a top on the No. 10 sieve. Gently tap the sieve assembly until no further material will pass through either sieve, as determined by observation. Weigh the material that was collected in the bottom pan to the nearest 0.05 gram. The weight of the sample shall be not less than 9.5 grams.

4.4.2.6.1 Calculation. Calculate the results as follows.

$$\text{Percent passing No. 20 sieve} = (\text{grams of material in pan})\ (10)$$

4.4.2.7 Bactericidal performance. The following procedure should be carried out on three separate 100 mg portions of the contents of the pouch A (type I or type II) to give three independent sets of results:

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a. Aseptically transfer to four sterilized 250 mL Erlenmeyer flasks 93 mL of cooled, sterilized hard water prepared as follows: To one liter of sterilized, cooled, glass distilled water, add 0.54 grams of sodium bicarbonate and 0.36 grams of anhydrous calcium chloride. CAUTION: Add these ingredients aseptically to the cooled sterilized water; hot water may cause decomposition of the bicarbonate.

b. Mark the flasks 1, 2, 3 and 4 and place in a 45° ± 1°C constant temperature water bath for 30 minutes or more. Weigh three 100 mg samples of the finished product of the pouch A (type I or type II) to the nearest 0.1 mg and introduce them separately into flasks 1, 2 and 3. Swirl the contents to facilitate solution and return the flasks to the bath.

c. After 10 minutes, pipette 2 mL of a freshly made, sterile 10 percent difco bacto-peptone solution or equal into each of the four flasks. Swirl the contents again and return to the bath.

d. After 120 minutes, pipette 5 mL of a suspension of bacterium Escherichia Coli (strain No. 26 ATCC available from the American Type Culture Collection, C/O Order Department, 12301 Parklawn Drive, Rockville, MD 20852. Telephone 1-800-638-6597 or (301) 881-2600) (see 4.4.2.7.1) containing approximately two billion cells per mL into each of the four flasks, using sterile pipettes, and mix thoroughly by continuous swirling in the bath.

e. After a contact of 90 seconds, pipette 2 mL of the final mixture in each of flasks 1, 2 and 3 separately into three sterile test tubes each containing 2 mL of a sterile 1 percent sodium sulfite (Na_2SO_3) solution (made up to contain 1.0 percent Na_2SO_3, 0.1 percent sodium pyro phosphate ($Na_4P_2O_7 10H_2O$), and 0.2 percent glycerine that is prepared no more than 24 hours before use. (NOTE: All glassware used in contact with the test fluid must be sterilized, and a fresh pipette must be used for each sample. Both the bacteria-dosing and the taking of sample must be timed so that the bacteria in each of the three test flasks shall have exactly 90 seconds of contact with the test fluid.)

f. Pour triplicate plates with 1 mL portions of each of the three neutralized samples before counting using tryptone glucose extract agar. Remove a 1 mL portion from flask 4 into 99 mL sterile water and mix thoroughly using trytone glucose extract agar. Make two more 1:100 serial dilutions in the same manner, and pour triplicate plates with 0.1 and 1.0 mL portions of the $1:10^6$ dilution using trytone glucose extract agar and incubate at 37° ± 1°C for 48 hours before counting. Multiply by two the counts on plates poured with the three test samples to correct for the dilution. Compute the percentage of survivors in each of the three test flasks from the average count per milliliter of each of the test samples and that of the control sample from flask 4.

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4.4.2.7.1 Inoculum. Wash off the surface growth on several 18 hour old nutrient-agar slants with sterile, normal saline, pooled and filtered through sterile No. 2 Whatman filter paper. Adjust the suspension to 2.0 billion cells per milliliter with a photelectric colorimeter or other suitable means. Store in the refrigerator until used. The suspension must be prepared fresh each day.

4.4.2.8 Formulation stability. Seal triplicate samples of the contents of the pouch A (type I or type II) of approximately 50 grams each (the titratable chlorine having been determined by the method specified in 4.4.2.1) in 4 ounce screwcap glass jars or other suitable glass containers and store for 60 days $\pm$ 4 hours in a suitable constant temperature oven maintained at 50° $\pm$ 1°C. At the end of the test period, determine the titratable chlorine of each sample by the method specified in 4.4.2.1. The average of the three samples shall be taken as the final value of titratable chlorine.

4.4.2.8.1 Calculation. Calculate the results as follows.

$$\text{Percent titratable chlorine remaining} = \frac{(\text{final value})\ (100)}{(\text{original value})}$$

4.4.2.9 Surface tension. Grind and mix thoroughly approximately 5 grams of the contents of the pouch A (type I or type II) in a glass mortar with a glass pestle. Weigh a 1 gram sample to the nearest 0.1 mg, obtained by sampling various portions of this finely ground and well mixed material, and dissolve in distilled water. Make up to 1 liter volume. Three suitable portions of this solution shall be used for determination of surface tension at 25°C using a calibrated du Nouty tensionmeter or a Roller Smith surface tension balance (platinum ring method). The average of the three determinations is obtained and is expressed as dynes per centimeter.

4.4.3 Tests on filled and sealed pouches.

4.4.3.1 Finished pouch stability. Place the finished pouch or pouches, as applicable, (filled and heat sealed) in a humidity cabinet for a period of 15 days $\pm$ 1 hour. Maintain the temperature and humidity range of the test cabinet at 38° $\pm$ 1°C with a minimum relative humidity of 90 percent.

4.4.3.2 Stability after packaging. Place the finished pouch or pouches, as applicable (filled and heat sealed), in a test cabinet maintained at 59° $\pm$ 1°C for a period of 15 days $\pm$ 1 hour.

4.4.3.3 Leakage of seams and closures of pouch. Place one filled and heat sealed pouch in a desiccator, or other suitable container, partially filled with water. Cover the pouch with a grating to prevent floating and to maintain a one inch height of water over the top surface of the filled pouch. Gradually apply a vacuum of 20 inches of mercury for a period of 3 minutes. Observe the pouch for a steady flow of air bubbles (leakage indicating) during this period.

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Isolated or static bubbles caused by entrapped air in the seams or on the surface of the pouch will not be considered as a sign of leak. Remove the pouch from the desiccator or other suitable container, dry hands and pouch and examine the inside of the pouch for presence of moisture.

4.4.3.4 Water spray test. Place the completed intermediate packs, filled and sealed as specified in 5.1.1, on a rack 1 inch in height and arrange side by side in the same position as they will be placed in the exterior shipping container. The spray shall consist of a spray of water covering the complete upper surfaces of the four intermediate packs. There shall be a minimum free fall of spray of 5 feet. The average fall of water shall be at the minimum rate of 2 gallons per square foot per hour. The spray shall be continuous for 4 hours ± 5 minutes.

PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Twelve bagged double-compartmented pouches of disinfectant, with an instruction sheet (see 3.5.2), shall be packed in a snug-fitting intermediate nontest corrugated fiberboard box; a folding paperboard box conforming to variety 1, style III, type G, class i of PPP-B-566; or a setup paperboard box conforming to type I, variety 1, class A, style 4 of PPP-B-676. Each fiberboard box shall be closed in accordance with the appendix of the applicable box document. Each box shall then be placed in a waterproof 0.002 inch thick polyethylene bag conforming to type I, grade A, style 1 of PPP-B-26. Heat sealed seams shall be a minimum of 1/16 inch in width. Prior to or during the heat sealed closure operation, excess air within the bag shall be expelled. The completed intermediate pack shall be water spray tested as specified in 4.4.3.4. After testing, there shall be no water or soggy or wet material within the intermediate pack.

5.1.2 Commercial. Disinfectant, with an instruction sheet (see 3.5.2), shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Ninety-six pouches (eight intermediate packs) of disinfectant, preserved as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to overseas type, style A or I, grade A or B, type 2 load of PPP-B-601. Level A intermediate packs shall be packed flat, two in length, two in width and two in depth within a shipping container. Each wood container shall be closed and reinforced in accordance with the appendix of PPP-B-601.

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5.2.2 Level B packing. Ninety-six pouches (eight intermediate packs) of disinfectant, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A intermediate packs shall be packed flat, two in length, two in width and two in depth within a shipping container. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant containers. When specified (see 6.2), the fiberboard shipping container shall be grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636.

5.2.3 Commercial packing. Disinfectant, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), disinfectant, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the bonding means K and L or film bonding O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, intermediate packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Special marking. Each intermediate pack shall have the following legend above the nomenclature:

This box contains 12-each double-compartmented pouches of (lot number).

5.4.2 Shelf-life marking. Each intermediate pack and shipping container shall be marked with the following shelf-life information located on same side and/or ends as the identification data in the order listed:

Type II Shelf-Life Item
Date of Manufacture _____
Expiration Date _____
Inspection/Test Date _____ Test No. _____
Extended/Reinspection Date _____

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5.4.2.1 Additional shelf-life marking. A label shall be affixed in accordance with MIL-STD-129 to each shipping container and the following legend shall be printed in white ink on red paper:

CAUTION

NOT TO BE USED AFTER _____ *

UNLESS CONTENTS YIELD A MINIMUM OF

17 PERCENT TITRATABLE CHLORINE WHEN

TESTED IN ACCORDANCE WITH INSTRUCTIONS

INSERTED IN INTERIOR PACKAGE

* Supplier shall insert the month and year that is 18 months after the date on the pack or date of last successful testing.

6. NOTES

6.1 Intended use. The food service disinfectant is intended for use in disinfection of fresh fruits and vegetables grown in areas where use of feces as fertilizer is suspected. The food service disinfectant is also intended for use in disinfection of mess gear and eating utensils. The food service disinfectant provides health protection to personnel against diseases such as amoebic dysentery, and under emergency conditions, it constitutes their sole protection. It is intended for use by troops and the amount packaged is sufficient for treating 25 gallons of water and 100 mess kits. The food service disinfectant covered by this document is intended for use only when hot water rinse is not available.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Type required (see 1.2).
c. When laboratory report is not required (see 3.1).
d. When weight of pouch required differs from specified weight (see 3.3.1.1 and 3.3.2.1).
e. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
g. When palletization is required (see 5.3).

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6.3 Component handling. Extreme care must be exercised in handling, grinding, mixing and packaging the components and formulation to protect against moisture and to ensure homogeniety of the formulation so that a consistently high quality product is produced that will remain stable during long time storage. All equipment should be kept clean and protected against moisture during any shutdown. During a shutdown of more than 12 hours, it is recommended that all adhering formulation be removed from the grinding, mixing and filling machines.

6.4 Specialized requirements. Contractor's attention is directed to the fact that this item is most difficult to produce and requires specialized equipment and technical know-how based on past experience to produce the superior product required. There can be no compromise or deviation from the high quality and stability standards set for this product. Components, all but for surface active material and potassium iodide, having the properties listed below have been used successfully in the past to prepare a food service disinfectant that conforms to this document.

6.4.1 Trichloromelamine. The trichloromelamine should be white to light cream in color, having a titratable chlorine of not less than 89.0 percent and no more than 94.5 percent, and a moisture content not greater than 0.5 percent.

6.4.1.1 Citric acid, anhydrous. The anhydrous citric acid should have a moisture content of not greater than 0.2 percent, and the assay show not less than 99.3 percent $C_6H_8O_7$.

6.4.1.2 Monosodium dihydrogen phosphate, anhydrous. The anhydrous monosodium dihydrogen phosphate should contain no less than 57.5 percent nor more than 59.1 percent of phosphorous pentoxide and should have a moisture content no greater than 2.0 percent.

6.4.2 Potassium dichloro-s-triazinetrione. Potassium dichloro-s-triazinetrione should be white in color, having a titratable chlorine of not less than 59.6 percent.

6.4.2.1 Tartaric acid, anhydrous. The anhydrous tartaric acid should have a moisture content of not greater than 0.2 percent, and the assay show not less than 99.7 percent $C_4H_6O_6$.

6.5 Grinding. In order to meet solubility requirements consistently, the procedures listed in 6.5.1 and 6.5.2 should be carried out.

6.5.1 Contents of trichloromelamine pouch. It is necessary to grind at least a portion of the finished formulation and all of the trichloromelamine and wetting agent in micronizing equipment. Trichloromelamine should not be ground alone or when mixed only with the monosodium dihydrogen phosphate because of fire hazard.

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6.5.2 Contents of chlorinated cyanurate. It is necessary to grind all of the ingredients other than chlorinated cyanurate in micronizing equipment. Powder or regular grade potassium dichloro-s-triazinetrione should be added in the end, when dry mixing all the ingredients in the formulation.

6.6 Heat sealable barrier material. The heat sealable barrier material specified in 3.3.1.1 and 3.3.1.2 is available by special order from Continental Can, Continental Group, Greenwich Office Park II, Greenwich, CT 06830.

6.7 Subject term (key word) listing.

Disinfectant
Food service

6.8 Changes from the previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Air Force - 68

Review activities:

Army - MD
Air Force - 03
DLA - GS

User activities:

Navy - MC, MS

Preparing activity:

Army - GL

Project No. 6840-0400

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DISINFECTANT, FOOD SERVICE (CHLORINE-IODINE TYPE)

6840-00-810-6396

MIL-D-11309G-TYPE I

For disinfection of mess gear, fresh fruits, and vegetables
(4.77 oz. net quantity for 25 gallons of water).

ACTIVE INGREDIENTS:

Trichloromelamine	19.3 %
Potassium Iodide	28.6 %
INERT INGREDIENTS	52.1 %

AVAILABLE CHLORINE	250 mg/l (ppm)
TITRATABLE IODINE	300 mg/l (ppm)

DANGER

SEE BACK PANEL FOR ADDITIONAL PRECAUTIONS AND FIRST AID

Name of Mfg & Address Lot No. Date

Contract No.

EPA Reg. No. 40510-1 EPA Est. No.

Inspection Test Date

Figure 1. EPA Approved Front Label

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DIRECTIONS FOR USE

IT IS A VIOLATION OF FEDERAL LAW TO USE THIS PRODUCT IN A MANNER INCONSISTENT WITH ITS LABELING

a. For disinfection of mess gear (intended for use only when hot water rinse is not available):

The double compartmented pouch is sufficient for treating 25 gallons of water resulting in a dosage of approximately 257 mg/l of chlorine and 300 mg/l of iodine. To treat 100 sets of mess gear, dissolve the contents of one double compartmented pouch of Disinfectant, Food Service (one trichloromelamine pouch labeled pouch A and one potassium iodide pouch labeled pouch B) in a container containing 25 gallons of water. Repeat the above procedure with a second container containing 25 gallons of rinse water. If water is cold, mix contents of the double compartmented pouch with 3 to 4 parts of water in a small container (canteen cup will do) and pour the mixture into the rinse water. Repeat for the second container of 25 gallons of water. Stir thoroughly to dissolve. After washing each set of mess gear, rinse twice- once in the first container of disinfectant solution and a final rinse in the second container of disinfectant solution. Allow mess gear to air dry. A new solution must be prepared after disinfecting 100 mess kits.

b. For fresh fruits and vegetables:

1. Remove bruised and torn leaves and damaged fruits or vegetables.
2. Do not cut or peel fruits or vegetables before disinfecting.
3. Wash thoroughly in a solution made by dissolving the contents of one double-compartmented pouch (pouch A and pouch B) in 20 gallons of water. This solution will yield approximately 320 mg/l of chlorine and 380 mg/l of iodine.
4. After washing, completely immerse the fruits and vegetables for ten minutes in a separate freshly prepared solution made by dissolving one double-compartmented pouch of Disinfectant, Food Service in 20 gallons of water. Stir occasionally to obtain thorough wetting of vegetables and fruit surfaces.
5. After ten minutes of immersion, remove the fruits and vegetables from the disinfectant solution and rinse thoroughly in potable water.

NOTE: SOLUTIONS OF DISINFECTANT, FOOD SERVICE ARE NOT TO BE REUSED, MAKE FRESH SOLUTIONS FOR EACH USE

STORAGE AND DISPOSAL

Store in a cool, dry, well ventilated area away from contact with other chemicals and solvents. Moisture contamination may liberate irritating and hazardous chlorine containing gases. Rinse empty pouches thoroughly with water and discard them in a safe place.

PRECAUTIONARY STATEMENTS HAZARDOUS TO HUMANS AND DOMESTIC ANIMALS

Harmful if swallowed. May cause eye damage. Do not get in eyes, on skin or clothing. Do not breathe dust.

If swallowed: Drink water, milk or egg whites. Call physician. In case of personal contact: Remove excess powder and flush skin and eyes with fresh water for at least 15 minutes. For eyes, get medical attention.

PHYSICAL OR CHEMICAL HAZARD

Contains trichloromelamine which may react vigorously to evolve heat and highly toxic smoke under the following conditions: If it is exposed to extreme heat, if ignited by spark or flame, if mixed before dilution in water with certain organic materials, acid and alkalis.

DO NOT MIX WITH OTHER CHEMICALS SUCH AS CLEANING COMPOUNDS, AMMONIA, ACIDS, SOLVENTS, OR OTHER ORGANIC MATERIAL.

Figure 2. EPA Approved Back Label

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DISINFECTANT, FOOD SERVICE (CHLORINE-IODINE TYPE)

MIL-D-11309G-TYPE II

For disinfection of mess gear, fresh fruits, and vegetables
(4.77 oz. net quantity for 25 gallons of water).

ACTIVE INGREDIENTS:

Potassium Dichloro-s-Triazinetrione	31.5 %
Potassium Iodide	28.6 %
INERT INGREDIENTS	39.9 %

AVAILABLE CHLORINE	0.0247% (247 ppm)
TITRATABLE IODINE	0.030% (300 ppm)

DANGER

SEE BACK PANEL FOR ADDITIONAL PRECAUTIONS AND FIRST AID

Name of Mfg & Address Lot No. Date

Contract No.

EPA Reg. No. EPA Est. No.

Inspection Test Date

Figure 3. EPA Approved Front Label

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DIRECTIONS FOR USE

IT IS A VIOLATION OF FEDERAL LAW TO USE THIS PRODUCT IN A MANNER INCONSISTENT WITH ITS LABELING

a. For disinfection of mess gear (intended for use only when hot water rinse is not available):

The double compartmented pouch is sufficient for treating 25 gallons of water resulting in a dosage of approximately 247 mg/l of chlorine and 300 mg/l of iodine. To treat 100 sets of mess gear, dissolve the contents of one double compartmented pouch of Disinfectant, Food Service (one potassium dichloro-s-triazinetrione pouch labeled pouch A and one potassium iodide pouch labeled pouch B) in a container containing 25 gallons of water. Repeat the above procedure with a second container containing 25 gallons of rinse water. If water is cold, mix contents of the double compartmented pouch with 3 to 4 parts of water in a small container (canteen cup will do) and pour the mixture into the rinse water. Repeat for the second container of 25 gallons of water. Stir thoroughly to dissolve. After washing each set of mess gear, rinse twice- once in the first container of disinfectant solution and a final rinse in the second container of disinfectant solution. Allow mess gear to air dry. A new solution must be prepared after disinfecting 100 mess kits.

b. For fresh fruits and vegetables:

1. Remove bruised and torn leaves and damaged fruits or vegetables.
2. Do not cut or peel fruits or vegetables before disinfecting.
3. Wash thoroughly in a solution made by dissolving the contents of one double-compartmented pouch (pouch A and pouch B) in 20 gallons of water. This solution will yield approximately 308 mg/l of chlorine and 380 mg/l of iodine.
4. After washing, completely immerse the fruits and vegetables for ten minutes in a separate freshly prepared solution made by dissolving one double-compartmented pouch of Disinfectant, Food Service in 20 gallons of water. Stir occasionally to obtain thorough wetting of vegetables and fruit surfaces.
5. After ten minutes of immersion, remove the fruits and vegetables from the disinfectant solution and rinse thoroughly in potable water.

NOTE: SOLUTIONS OF DISINFECTANT, FOOD SERVICE ARE NOT TO BE REUSED, MAKE FRESH SOLUTIONS FOR EACH USE

STORAGE AND DISPOSAL

Store in a cool, dry, well ventilated area away from contact with other chemicals and solvents. Moisture contamination may liberate irritating and hazardous chlorine containing gases. Rinse empty pouches thoroughly with water and discard them in a safe place.

PRECAUTIONARY STATEMENTS HAZARDOUS TO HUMANS AND DOMESTIC ANIMALS

Harmful if swallowed. May cause eye damage. Do not get in eyes, on skin or clothing. Do not breathe dust.

If swallowed: Drink water, milk or egg whites. Call physician. In case of personal contact: Remove excess powder and flush skin and eyes with fresh water for at least 15 minutes. For eyes, get medical attention.

PHYSICAL OR CHEMICAL HAZARD

Contains potassium dichloro-s-triazinetrione which may react vigorously to evolve heat and highly toxic smoke under the following conditions: If it is exposed to extreme heat, if ignited by spark or flame, if mixed before dilution in water with certain organic materials, acid and alkalis.

DO NOT MIX WITH OTHER CHEMICALS SUCH AS CLEANING COMPOUNDS, AMMONIA, ACIDS, SOLVENTS, OR OTHER ORGANIC MATERIAL.

Figure 4. EPA Approved Back Label

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-11309G	DISINFECTANT, FOOD SERVICE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 20 Nov 84